Weather-to-Satellite Drought Early Warning for Rayalaseema

Final results of stages 1-2: weather forecasting and satellite confirmation

Project	DeepDroughtUAV: drought-area estimation in the Anantapur cluster
Prepared by	Aryan Rajesh Gadam
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Study area	8 farming areas in Sri Sathya Sai and Kurnool (Andhra Pradesh) and Tumakuru (Karnataka)
Code and data	github.com/ARYAN3012-G/weather_time_series github.com/ARYAN3012-G/Sentinel2-Drought-Confirmation

1. Summary

A weather-to-satellite early-warning chain for rainfed farmland confirmed every severe drought season declared in 2023-2026 (9 of 9, Andhra Pradesh and Karnataka) and stayed silent in 10 of 11 seasons with no declaration. All results are on held-out years, with leakage tests, physical and statistical baselines and confidence intervals.

Key results

- **Weather stage:** drought class 7 days ahead with 85.6% balanced accuracy (68.3% at 15 days); macro-F1 0.819 / 0.651 vs 0.662 / 0.515 for persistence.
- **Physics matters:** a water-balance ensemble alone reaches 0.783 at 7 days; machine learning adds significant skill at 15 days; nothing is skilful at 30 days.
- **Crops follow weather:** rainfed crop greenness (SVI) follows 30-day SPEI 10 days earlier (ρ 0.48 in training, 0.45 on unseen years 2024-2026).
- **Early warning:** fields about to fall below normal within 15 days are ranked at ROC-AUC 0.76, +0.02 to +0.07 over a fair statistical baseline.
- **Operational replay:** severe 9/9 confirmed, 10/11 undeclared seasons quiet, 8 of 15 declared seasons confirmed before the government order (median 7 days earlier).
- **Honest negatives:** weather adds no significant skill to crop forecasts beyond satellite history; single-date agreement with UAV verdicts is weak (3 of 10 fields clear, 2 mixed).

2. Study design

The weather alarm says where to look, the satellite says which fields were actually hit, and the drone measures the damaged area. This report covers stages 1-2; stage 3 is the UAV segmentation work of the DeepDroughtUAV team.

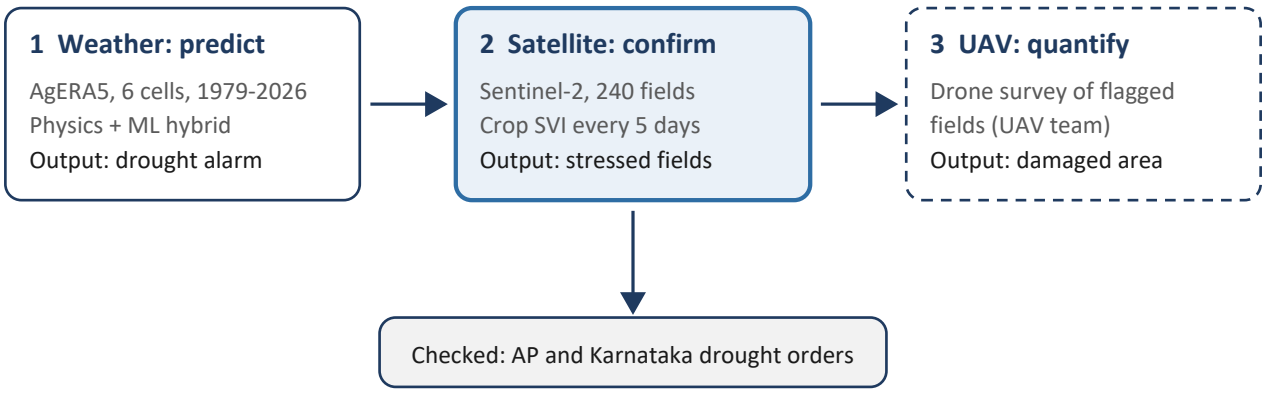


Figure 1. The three-stage system.

Each site centre is a UAV survey field; the satellite uses farmland within 3 km of it. Names and mandals were checked against the coordinates.

Site	Village	Mandal (district)	Centre (°N, °E)	Weather cell
Somandepalle	Tungodu	Somandepalle (Sri Sathya Sai)	13.9988, 77.6247	14.0, 77.6
Lepakshi	Mamidimakalapalli	Lepakshi (Sri Sathya Sai)	13.7998, 77.6194	13.8, 77.6*
Cholasamudram	Cholasamudram	Lepakshi (Sri Sathya Sai)	13.8118, 77.5687	13.8, 77.6*
Sira	Gidaganahalli	Sira taluk (Tumakuru, KA)	13.7975, 76.9750	13.8, 77.0
Peddakadubur	Pedda Thumbalam	Adoni / Peddakadubur (Kurnool)	15.7826, 77.2528	15.8, 77.3
Kurnool - Dinnedevarapadu	Dinnedevarapadu	Kurnool Rural	15.7715, 78.0471	15.8, 78.0
Kurnool - Balaji Villas	Kurnool outskirts	Kurnool Urban	15.7834, 78.0561	15.8, 78.1*
Kurnool - Pullaiah College	Venkayyapalle	Kurnool Urban	15.7982, 78.0806	15.8, 78.1*

*Table 1. Sites. * = weather cell shared by two sites (6 independent cells). Periods: weather training 1979-2011, validation 2012-2018, test 2019-2026; satellite training 2019-2023, validation 2024, test 2025-2026.*

3. Data and official records

Source	Content	Period	Use
AgERA5 v2.0 (Copernicus CDS)	Daily rain, FAO-56 ET0, temperature, humidity, radiation, wind (0.1° cell of each site)	1979 - Nov 2026	Weather inputs, SPEI
Sentinel-2 SR (Earth Engine)	B2-B12, NDVI, NDMI, EVI per field and date; Cloud Score+ and SCL masks	2019 - Oct 2026	Crop condition (SVI)
ESA WorldCover, Dynamic World	10 m land cover	2019-2023	Finding farmland
UAV dataset (DeepDroughtUAV)	10 surveyed fields, 2,254 images, drought / healthy labels	Nov 2025 - Jan 2026	Field check
AP G.O.Ms. 4/2023, 15/2024, 6/2025, 4 & 5/2026; Karnataka 2023 order	Drought mandal / taluk lists, severe / moderate	Kharif 2023-2026	Independent check

Table 2. Data. All inputs are public; nothing is simulated.

4. Stage 1: weather module

The module forecasts the SPEI-30 drought class (normal ≥ 0, moderate -1.5 to 0, severe < -1.5) 7, 15 and 30 days ahead. SPEI is calibrated on 1979-2011 only and frozen; the split is chronological and purged, and models are chosen on validation data. The **water-balance ensemble** is a physical baseline: on forecast day t part of the 30-day window is already observed, and the unknown days are filled with each 1979-2011 year's water balance in turn. The **hybrid** adds its three class probabilities to the weather, SPEI and season inputs.

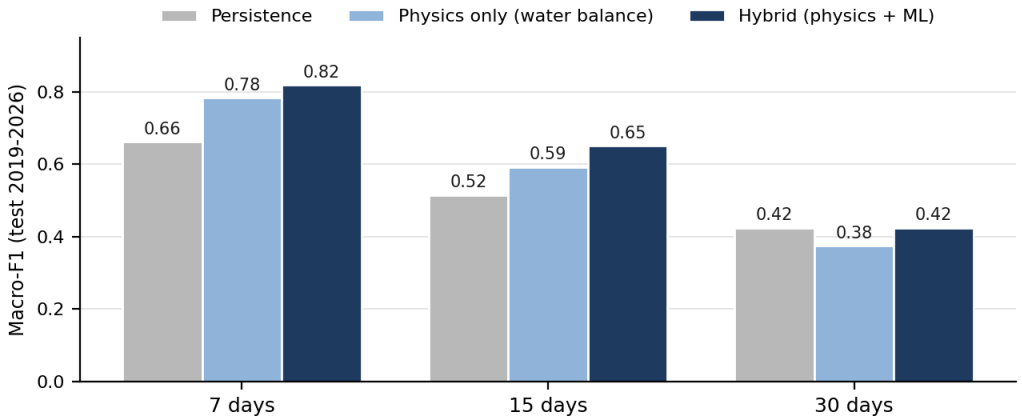


Figure 2. Weather-module test scores, all sites.

Model	7 days	15 days	30 days
Persistence (no change)	0.662	0.515	0.424
Water-balance ensemble (physics only)	0.783	0.592	0.375
Best model without physics (XGBoost)	0.755	0.613	0.424
Hybrid, ensemble of 4 models	**0.819**	**0.651**	**0.424**
Hybrid vs physics, Δ [95% CI]	[-0.007, +0.088]	[+0.010, +0.099]	[+0.023, +0.084]
Balanced accuracy, hybrid	85.6%	68.3%	42.5%
ROC-AUC, hybrid	0.97	0.90	0.68

Table 3. Macro-F1 on test years 2019-2026 unless stated. 30 days: no model beats persistence.

- **7 days:** physics is hard to beat; the hybrid gain over it is not significant, but both beat persistence widely.
- **15 days:** the hybrid significantly beats physics and persistence.
- **Unseen regions** (leave-one-cluster-out): the hybrid holds up (mean 0.811 / 0.633 / 0.381).
- **Marusov et al. protocol** (binary SPEI < -1): weather inputs raise 15-day ROC-AUC from 0.81 to 0.89; at 30 days a logistic model on the last SPEI is as good as anything.

5. Stage 2: satellite module

Farmland is found automatically in Earth Engine from training years 2019-2023 only: ESA WorldCover cropland, minus built-up, tree and water pixels (Dynamic World), with a crop cycle in at least 2 of 5 years. SNIC splits it into 13,750 field segments; 20 rainfed and 10 irrigated fields are sampled per site (240 fields). The label is the **Standardized Vegetation Index** (SVI = NDVI departure from the field's own normal / its spread), with normals from images up to 31 Dec 2023 and the scored year fully removed.

Leakage controls, each covered by automated tests (42 across both modules):

- model inputs use only images up to the forecast date; spike cleaning (which looks ahead) is applied to labels only;
- normals never see later years or the scored year; samples are purged when inputs, target or label support cross a period boundary;
- an independent code review found five look-ahead and selection issues in an earlier version; all are fixed and every result here is from the corrected pipeline.

6. Result: crops follow weather drought within about 10 days

The SPEI scale and lag were chosen on 2019-2023 only, then measured on 2024-2026.

Fields	Chosen on 2019-2023	ρ training	ρ 2024-2026 (median, range)
Rainfed	SPEI-30, 10-day lag	0.48	0.45 (0.16-0.58)
Irrigated	SPEI-60, 15-day lag	0.24	0.29 (0.12-0.46)

Table 4. Spearman correlation of crop SVI with SPEI. Rainfed minus irrigated on 2024-2026: [-0.02, +0.24].

Weather alarms (SPEI-30 below -1) were followed by rainfed crop stress within 60 days in 22 of 44 cases, clearly above the base rate at Balaji Villas, Dinnedevarapadu and Cholasamudram.

7. Result: crop forecasts and early warning of stress

Version	15 days	Δ vs persistence [95% CI]	30 days
Persistence (no change)	0.648		0.520
1 Satellite history	0.668	[-0.004, +0.036]	0.555
2 + past weather	0.677	[+0.011, +0.038]	0.553
3 + weather-module forecast	0.677	[+0.013, +0.041]	0.553
4 + actual future weather (upper bound)	0.685	[+0.016, +0.052]	0.575

Table 5. Macro-F1 on test years 2025-2026 (XGBoost, 5 seeds).

- Weather versions beat persistence at 15 days (also with 90-day bootstrap blocks) but not satellite history alone; the weather forecast adds nothing beyond past weather.
- **Onset warning**: among fields normal today, the models rank those falling below normal by day 15 at ROC-AUC 0.76 (30 days: 0.70), vs 0.72 for a logistic baseline on current SVI and its trend.
- Interpretation: crop stress at 15-30 days is set mostly by water already lost, so the weather stage's value is the earlier meteorological alarm.

8. Result: operational replay against government declarations

Every 5 days from 1 June to 30 November, with weather 7 days late and satellite 2 days late. Alarm: SPEI-30 < -1 or a severe +15-day forecast. Confirmed: median rainfed SVI < -1 from at least 5 clear fields; fewer fields = 'insufficient data'. No crop model is used, so all seasons are out of sample.

Site	2023	2024	2025	2026
Somandepalle	Moderate: 2 d before	none: quiet	Moderate: missed	Severe: after order
Lepakshi	Moderate: 7 d before	none: quiet	Moderate: missed	Severe: 7 d before
Cholasamudram	Moderate: 12 d before	none: quiet	Moderate: missed	Severe: after order
Sira (Karnataka)	Severe: 5 d before	none found	none found	likely declared
Kurnool - Balaji Villas	Severe: 2 d before	none: quiet	none: quiet	Severe: 22 d before
Kurnool - Dinnedevarapadu	none: confirmed	none: quiet	none: quiet	Severe: 22 d before
Kurnool - Pullaiah	Severe: after order	none: quiet	none: quiet	Severe: after order
Peddakadubur	ambiguous mandal	ambiguous	none: quiet	ambiguous

Table 6. Official declaration and system result per site and season.

Official declaration	System confirmed	Not confirmed
Severe (9 site-seasons)	9	0
Moderate (6)	3	3 (all 2025)
Not declared (11)	1	10

Table 7. Agreement. 8 of 15 declared site-seasons were confirmed before the order (median 7 days). Thresholds -0.75 / -1.25 leave the false-alarm count unchanged.

9. Mismatches and the UAV fields

Case	Observation	Likely reason (to verify)
2025: Lepakshi, Somandepalli declared Moderate	Rain above normal; crops normal or greener June-October	Late-season stress from November-December, after the replay window
2023: Dinnedevarapadu not declared	53% of fields below normal	Mandal-level assessment vs a 3 km urban-fringe area
2026: Peddakadubur	Driest season (-57% rain), only 25% of fields below normal	Deep black soils may hold water longer (check soil map)
UAV fields	Clear agreement at 3 of 10 fields, 2 mixed	Post-harvest flights; drone sees mm patches, Sentinel-2 averages 10 m

Table 8. Cases that do not fit. At the southern 'drought' maize fields (15 Nov 2025) Sentinel-2 showed crops greener than normal in September-October and below normal only from December: a field with 40% dry patches can look normal at 10 m. This is why the UAV stage is needed after satellite confirmation.

10. What is new

Aspect	Common practice	This work
Baselines	Other ML models	Persistence, Markov, climatology and a physical water-balance ensemble
Calibration	Full record	SPEI frozen on 1979-2011; SVI normals on 2019-2023, scored year removed
Leakage	Rarely tested	42 automated tests, independent review, purging by label support
Fields	Hand-picked or district averages	Automatic farmland from training years; 240 random fields
Validation	Index vs index	Four seasons of government orders, live replay with data delays, UAV fields
Weather + satellite	Studied separately	Linked; direct test of whether forecasts improve crop forecasts

Table 9. Comparison with common practice. The Marusov et al. comparison is a Marusov-inspired protocol, not a claim of beating their monthly PDSI results.

11. Limitations

- Short satellite record: normals rest on five years and the test years hold one full drought season (2026).
- Few events: 3-7 alarms per site and 15 declared site-seasons; agreement figures are case evidence.
- Monsoon clouds often leave fewer than 5 clear fields in August-September, delaying confirmation.
- SVI classes are index categories, not measured damage; irrigation is inferred from dry-season greenness.
- Spatial mismatch between mandals, 3 km areas and single fields; for Sira only 2023 is confirmed by name.
- AgERA5 is about a week late, so 7-day forecasts give little real warning; 15 days gives about a week.
- No skill beyond persistence at 30 days; UAV comparison uses post-harvest flights and no field boundaries.

12. References

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